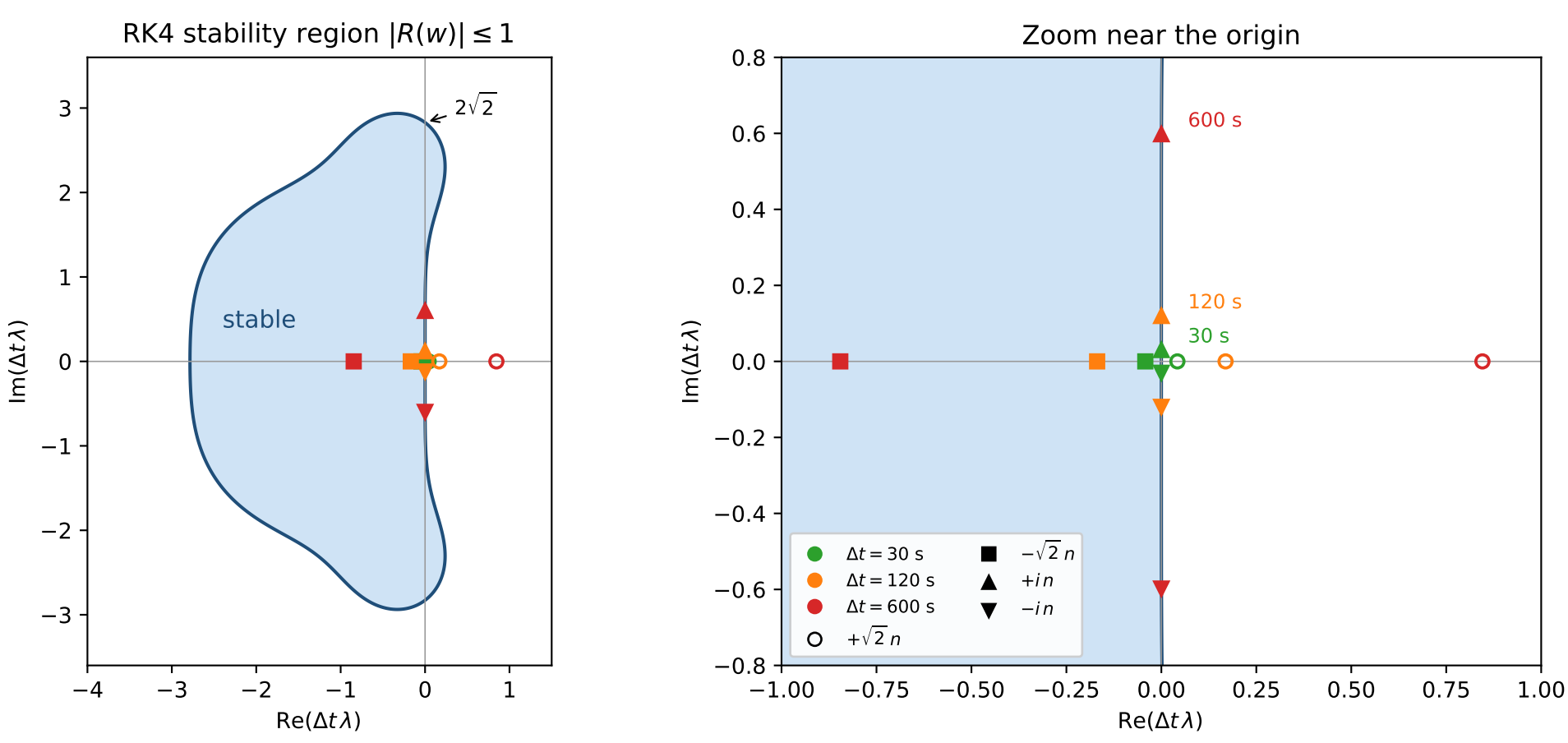

RK4 stability region $|R(w)| \leq 1$
stable
$2\sqrt{2}$
$\mathrm{Im}(\Delta t\lambda)$
$\mathrm{Re}(\Delta t\lambda)$
Zoom near the origin
600 s
120 s
30 s
$\Delta t = 30$ s
$\Delta t = 120$ s
$\Delta t = 600$ s
$+\sqrt{2}\,n$
$-\sqrt{2}\,n$
$+in$
$-in$